SPEECH EMOTION DETECTOR

A PROJECT REPORT

Submitted by

ISHWARYA A(8115U24AM025)

in partial fulfilment for the award of the course

AMB 1221-MACHINE LEARNING FUNDAMENTALS

IN

DEPARTMENT OF

COMPUTER SCIENCE AND ENGINEERING

(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)
SAMAYAPURAM, TRICHY

ANNA UNIVERSITY
CHENNAI 600 025

MAY 2026

SPEECH EMOTION DETECTOR

A PROJECT REPORT

Submitted by

ISHWARYA A (8115U24AM025)

in partial fulfilment for the award of the course

AMB1221-MACHINE LEARNING FUNDAMENTALS

IN

DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

Under the Guidance of

Mrs. C.MUTHUSELVI

Department of Artificial Intelligence and Data Science
K. RAMAKRISHNAN COLLEGE OF ENGINEERING

K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)

ANNA UNIVERSITY, CHENNAI

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

(AUTONOMOUS)

ANNA UNIVERSITY, CHENNAI

BONAFIDE CERTIFICATE

Certified that this project report titled **"SPEECH EMOTION DETECTOR"** is the Bonafide work of **ISHWARYA A (8115U24AM025)** who carried out the work under my supervision.

SIGNATURE

Dr. B.KIRAN BALA, M.E.,M.B.A.,Ph.D.,

HEAD OF THE DEPARTMENT,
ASSOCIATE PROFESSOR,
Department of Artificial Intelligence and Machine Learning,
K. Ramakrishnan College of Engineering, (Autonomous)
Samayapuram, Trichy.

SIGNATURE

Mrs. C. MUTHUSELVI M.E.,

SUPERVISOR,
ASSISTANT PROFESSOR,
Department of Artificial Intelligence and Machine Learning
K. Ramakrishnan College of Engineering, (Autonomous)
Samayapuram, Trichy.

SIGNATURE OF INTERNAL EXAMINER

NAME:
DATE:

SIGNATURE OF EXTERNAL EXAMINER

NAME:
DATE:

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

(AUTONOMOUS)

ANNA UNIVERSITY, CHENNAI

DECLARATION BY THE CANDIDATE

I declare that to the best of my knowledge the work reported here in has been composed solely by myself and that it has not been in whole or in part in any previous application for a degree.

Submitted for the project Viva-Voice held at K. Ramakrishnan College of Engineering on _____

SIGNATURE OF THE CANDIDATE

ACKNOWLEDGEMENT

I thank the almighty GOD, without whom it would not have been possible for me to complete my project.

I wish to address my profound gratitude to **Dr.K.RAMAKRISHNAN**, Chairman, K. Ramakrishnan College of Engineering(Autonomous), who encouraged and gave me all help throughout the course.

I extend my hearty gratitude and thanks to my honorable and grateful Executive Director **Dr.S.KUPPUSAMY, B.Sc., MBA., Ph.D.,** K. Ramakrishnan College of Engineering(Autonomous).

I am glad to thank my Principal **Dr.D.SRINIVASAN, M.E., Ph.D.,FIE., MIIW., MISTE., MISAE., C.Engg,** for giving me permission to carry out this project.

I wish to convey my sincere thanks to **Dr.B.KIRAN BALA, M.E., M.B.A., Ph.D.,** Head of the Department, Artificial Intelligence and Data Science for giving me constant encouragement and advice throughout the course.

I am grateful to **Mrs. C.MUTHUSELVI, M.E.,** Assistant Professor, Artificial Intelligence and Data Science, K. Ramakrishnan College of Engineering (Autonomous), for her guidance and valuable suggestions during the course of study.

Finally, I sincerely acknowledged in no less terms all my staff members, my parents and, friends for their co-operation and help at various stages of this project work.

ISHWARYA A (8115U24AM025)

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

(AUTONOMOUS)

ANNA UNIVERSITY, CHENNAI

DEPARTMENT OF CSE (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

VISION

To become a renowned hub for AIML technologies to producing highly talented globally recognizable technocrats to meet industrial needs and societal expectation.

MISSION

Mission of the Department

M1 To impart advanced education in AI and Machine Learning, built upon a foundation in Computer Science and Engineering.

M2 To foster Experiential learning equips students with engineering skills to tackle real-world problems.

M3 To promote collaborative innovation in AI, machine learning, and related research and development with industries.

M4 To provide an enjoyable environment for pursuing excellence while upholding strong personal and professional values and ethics.

PROGRAM EDUCATIONAL OBJECTIVES (PEO's)

PEO1 Excel in technical abilities to build intelligent systems in the fields of AI & ML in order to find new opportunities.

PEO2 Embrace new technology to solve real-world problems, whether alone or as a team, while prioritizing ethics and societal benefits.

PEO3 Accept lifelong learning to expand future opportunities in research and product development.

PROGRAM SPECIFIC OUTCOMES (PSO's)

PSO1 Expertise in tailoring ML algorithms and models to excel in designated applications and fields.

PSO2 Ability to conduct research, contributing to machine learning advancements and innovations that tackle emerging societal challenges.

PROGRAM OUTCOMES (POs)

Engineering students will be able to:

PO1 Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2 Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3 Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4 Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5 Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO6 The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7 Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO8 Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings

PO9 Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO10 Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO11 Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

ABSTRACT

This project presents the design and development of a **Speech Emotion Detector** System using Machine Learning techniques. The main objective of the system is to analyze human speech and automatically identify the emotional state of a person, such as happy, sad, angry, calm, fear, or neutral. The system processes both pre-recorded audio files and real-time voice input through a web-based interface.The proposed system utilizes audio signal processing techniques to extract important features such as Mel Frequency Cepstral Coefficients (MFCC) from speech data. These features are then used to train a Machine Learning model, specifically a Multi-Layer Perceptron (MLP) classifier, which learns to classify different emotional states based on patterns in the audio signals. The model is trained using a standard emotional speech dataset to ensure accurate and reliable predictions. A user-friendly web application is developed using Flask, which allows users to upload audio files or record their voice in real time using a microphone. The recorded or uploaded audio is processed by the trained model, and the predicted emotion is displayed along with relevant emojis for better user experience. The system is designed to provide quick responses and an interactive interface.This project demonstrates the practical application of machine learning and speech processing in real-world scenarios such as human-computer interaction, mental health analysis, customer service monitoring, and virtual assistants. The system aims to enhance communication between humans and machines by enabling emotion-aware interactions.

ABSTRACT WITH POs AND PSOs MAPPING

ABSTRACT	POs MAPPED	PSOs MAPPED
This project focuses on the development of a **Speech Emotion Detector** using machine learning techniques to identify human emotions from voice signals. The system analyzes both uploaded audio files and real-time recorded speech through a web-based application. Audio features such as Mel Frequency Cepstral Coefficients (MFCC) are extracted and used to train a classification model, which predicts emotions like happy, sad, angry, calm, and neutral. The application is built using Python and Flask, providing an interactive interface where users can record or upload audio and receive instant emotion predictions along with emojis. This system has practical applications in areas such as human-computer interaction, mental health monitoring, and customer service analysis, enabling more intelligent and emotion-aware communication systems.	**PO1-3** **PO2-3** **PO3-3** **PO4-3** **PO5-3** **PO6-3** **PO7-3** **PO8-3** **PO9-3** **PO10-3** **PO11-3**	**PSO1 – 3** **PSO2 – 3**

Note: 1- Low, 2-Medium, 3- High

TABLE OF CONTENTS

C HAPTER No.	TITLE	PAGE No.
	ABSTRACT	**viii**
	LIST OF ABBREVIATIONS	**xii**
	LIST OF FIGURES	**xiii**
1	**INTRODUCTION**	**1**
	1.1 Objective	1
	1.2 Overview	1
	1.3 Purpose and Importance	2
	1.4 Data Source Description	2
	1.5 Project Summarization	3
2	**LITERATURE SURVEY**	**4**
	2.1 Introduction and Traditional Methods	4
	2.2 Machine Learning Techniques for Emotion Detection	4
	2.3 Web-Based Systems and Visualization Techniques	5
	2.4 Limitations of Existing Systems and Summary	5
3	**PROJECT METHODOLOGY**	**6**
	3.1 Proposed Workflow	6
	3.2 Architectural Overview	7
	3.3 Hardware and Software Requirements	8
	3.4 System Integration and Working	8

4	**RELEVANCE OF THE PROJECT**	**9**
	4.1 Why the Model Was Chosen	9
	4.2 Comparison with Other Models	9
	4.3 Advantages and Disadvantages	10
	4.4 Summary of Relevance	11
5	**MODULE DESCRIPTION**	**12**
	5.1 Data Collection Module	12
	5.2 Data Preprocessing Module	12
	5.3 Feature Extraction Module	13
	5.4 Model Training Module	13
	5.5 Prediction Module	14
	5.6 Web Interface Module	15
	5.7 Evaluation Module	15
6	**RESULT AND DISCUSSION**	**16**
	6.1 Performance Analysis	16
	6.2 User Feedback	16
7	**CONCLUSION AND FUTURE WORK**	**18**
	7.1 Summary of Outcomes	18
	7.2 Future Scope and Enhancements	18
8	**APPENDIX**	**20**
	APPENDIXA – Source Code	20
	APPENDIXB – Screenshots	24
	REFERENCES	**26**

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE NO.
3.2.1	System Architecture	7
5.4.1	Multi-Layer Perceptron (MLP) Model	14
B.1	Home Page	24
B.2	File Uploading Screen	24
B.3	Prediction Results Output	25

LIST OF ABBREVIATIONS

S. NO.	ACRONYM	ABBREVIATION
1	AI	Artificial Intelligence
2	API	Application Programming Interface
3	CSS	Cascading Style Sheets
4	CSV	Comma Separated Values
5	HTML	HyperText Markup Language
6	HTTP	HyperText Transfer Protocol
7	KNN	K-Nearest Neighbors
8	MFCC	Mel Frequency Cepstral Coefficients
9	ML	Machine Learning
10	MLP	Multi-Layer Perceptron
11	RNN	Recurrent Neural Network
12	SVM	Support Vector Machine
13	URL	Uniform Resource Locator
14	WAV	Waveform Audio File Format

CHAPTER 1

INTRODUCTION

1.1 Objective

The primary objective of this project is to design and develop an intelligent and efficient system that can automatically detect and classify human emotions from speech signals using machine learning techniques. The system aims to process both pre-recorded and real-time audio inputs, analyze the speech data, and extract significant acoustic features such as Mel Frequency Cepstral Coefficients (MFCC), which are essential for identifying emotional characteristics in voice signals. Another key objective is to build and train a robust classification model capable of accurately recognizing a wide range of emotions including happy, sad, angry, calm, fear, disgust, surprise, and neutral, based on patterns learned from standard emotional speech datasets. In addition, the project focuses on developing a user-friendly and interactive web-based application using Flask, which enables users to easily upload audio files or record their voice through a microphone for real-time emotion detection. The system is designed to deliver quick and reliable predictions, displaying the detected emotion along with intuitive visual cues such as emojis to enhance user experience and understanding. Furthermore, the project aims to improve the interaction between humans and computers by making systems more responsive to human emotions, thereby contributing to more natural and effective communication.

1.2 Overview

The Speech Emotion Detection System is a machine learning–based application designed to identify human emotions from speech signals. The project focuses on analyzing voice input, either through uploaded audio files or real-time recording, and classifying the emotional state of the speaker into categories such as happy, sad, angry, calm, fear, disgust, surprise, and neutral. The system combines speech processing techniques with machine learning algorithms to recognize patterns in audio data and produce accurate emotion predictions.

The web application provides an interactive interface where users can upload audio files or record their voice using a microphone. The input audio is processed in real time, passed through the trained model, and the predicted emotion is displayed on the screen along with visual indicators like emojis. The system is designed to be user-friendly, responsive, and efficient, making it suitable for practical use.

1.3 Purpose and Importance

The purpose of this project is to design and develop a Speech Emotion Detection System that can automatically identify and classify human emotions from speech signals using machine learning techniques. The system is capable of processing both pre-recorded audio files and real-time voice input, analyzing the speech, and predicting emotions such as happy, sad, angry, calm, and neutral. It also aims to provide an interactive and user-friendly web-based platform where users can easily upload or record audio and receive instant emotion predictions along with visual representations like emojis.The importance of this project lies in its ability to enhance human-computer interaction by enabling machines to understand and respond to human emotions more effectively. In today's technology-driven world, most systems lack emotional awareness, and this project helps bridge that gap by introducing emotion recognition capabilities. It has wide applications in areas such as mental health monitoring, where emotional analysis can help detect stress or depression, customer service evaluation through voice-based feedback analysis, virtual assistants that respond more intelligently, and smart systems that adapt based on user emotions. Overall, this project demonstrates the practical implementation of machine learning and speech processing in creating intelligent, responsive, and emotion-aware systems.

1.4 Data Source Description

The data used in this project is obtained from the RAVDESS, which is a widely used and reliable dataset for speech emotion recognition. This dataset contains high-quality audio recordings of human speech expressed with different emotions such as happy, sad, angry, calm, fear, disgust, surprise, and neutral. The recordings are performed by professional actors under controlled conditions, ensuring clarity and consistency in the data.

Each audio file in the dataset follows a structured naming convention that encodes information about the emotion, intensity, and speaker, making it easy to organize and preprocess the data. The audio files are in .wav format and are pre-labeled, which is suitable for supervised machine learning models. In this project, the dataset is used to train and test the emotion classification model by extracting important features such as Mel Frequency Cepstral Coefficients (MFCC).

The use of the RAVDESS dataset improves the accuracy and reliability of the system, as it provides diverse and well-structured emotional speech samples. It enables the model to learn effectively and generalize better for real-time emotion prediction from user input audio.

1.5 Project Summarization

The Speech Emotion Detection System is a machine learning–based application developed to identify human emotions from speech signals. The project focuses on analyzing both pre-recorded audio files and real-time voice input to detect emotions such as happy, sad, angry, calm, fear, disgust, surprise, and neutral. It combines speech processing techniques with machine learning algorithms to accurately classify emotional states based on patterns present in voice data.The system works by collecting an emotional speech dataset and extracting important audio features such as Mel Frequency Cepstral Coefficients (MFCC). These features are then used to train a classification model, such as a Multi-Layer Perceptron (MLP), which learns to distinguish between different emotions. Once the model is trained, it is integrated into a web application developed using Flask, providing an interactive interface for users.
Users can upload audio files or record their voice in real time using a microphone, and the system processes the input to predict the corresponding emotion. The predicted result is displayed on the web page along with emojis to enhance user experience. The application is designed to be simple, efficient, and responsive, ensuring quick processing and accurate results.

Overall, this project demonstrates the effective use of machine learning and speech analysis to create an intelligent system capable of understanding human emotions. It has practical applications in areas such as human-computer interaction, mental health monitoring, customer service analysis, and virtual assistants, and can be further improved with advanced techniques and larger datasets for better performance

CHAPTER 2
LITERATURE SURVEY

2.1 Introduction and Traditional Methods

Speech Emotion Detection is an important area of research in the field of machine learning and speech processing, which focuses on identifying human emotions from voice signals. Human speech carries not only linguistic information but also emotional cues that reflect the speaker's state of mind. With the rapid growth of artificial intelligence and human-computer interaction systems, there is a growing need for machines to understand and interpret these emotional aspects of speech. This has led to the development of various techniques and models that aim to automatically recognize emotions such as happiness, sadness, anger, fear, and neutrality from audio data. The literature survey explores the existing methods, techniques, and approaches that have been proposed by researchers in the field of speech emotion recognition, highlighting their advantages, limitations, and performance.

2.2 Machine Learning Techniques for Emotion Detection

Machine learning techniques play a vital role in emotion detection by enabling systems to learn patterns from speech data and accurately classify different emotional states. In speech emotion recognition, important features such as Mel Frequency Cepstral Coefficients (MFCC), pitch, energy, and spectral characteristics are extracted from audio signals and used as input for various machine learning models. Algorithms like Support Vector Machines (SVM), k-Nearest Neighbors (KNN), Decision Trees, Random Forest, and Artificial Neural Networks (ANN), especially Multi-Layer Perceptrons (MLP), are commonly used for classifying emotions such as happy, sad, angry, calm, and neutral. These models are trained using labeled emotional speech datasets to improve prediction accuracy. In addition, advanced deep learning techniques like Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN), including LSTM, have further enhanced emotion detection by automatically learning complex patterns from audio data. Overall, machine learning provides an effective and flexible approach for developing accurate and intelligent speech emotion detection systems.

2.3 Web-Based Systems and Visualization Techniques

Web-based systems and visualization techniques play a significant role in making the Speech Emotion Detection System interactive, efficient, and user-friendly. In this project, a web-based interface is developed using Flask for the backend processing, while HTML and CSS are used to design an attractive and responsive user interface. The system enables users to upload audio files or record their voice in real time directly through a web browser, eliminating the need for complex software installation. Once the audio input is received, it is processed by the machine learning model to predict the corresponding emotion. The predicted result is then displayed on the web page using simple and effective visualization techniques such as text labels and emojis, which help users easily understand the emotional state. Additionally, the interface is designed to provide quick responses and smooth interaction, enhancing the overall user experience. The use of Flask ensures efficient communication between the frontend and backend, while HTML and CSS improve the layout and visual appeal of the application. These technologies together make the system accessible, easy to operate, and suitable for real-time applications, thereby increasing its practical usability in various domains.

2.4 Limitations of Existing Systems and Summary

Existing systems for speech emotion detection face several limitations that affect their overall performance and reliability. Many traditional methods rely heavily on manual feature extraction, which requires domain expertise and may not capture all relevant emotional patterns in speech. These systems often struggle with variations in voice such as different accents, speaking styles, background noise, and recording conditions, leading to reduced accuracy. Real-time emotion detection can also be challenging due to processing delays and computational complexity. Moreover, most existing systems lack proper user-friendly interfaces and effective visualization, making them less interactive and harder to use. In summary, while significant progress has been made in the field of speech emotion recognition, there are still challenges in achieving high accuracy, real-time performance, and robustness across diverse conditions. This project addresses some of these limitations by using machine learning techniques, standard datasets, and a web-based interface to provide a more efficient, interactive, and practical solution for emotion detection from speech.

CHAPTER 3
PROJECT METHODOLOGY

The methodology of the Speech Emotion Detector follows a structured process starting with the collection of a standard emotional speech dataset such as RAVDESS, which contains labeled audio samples of different emotions. The audio data is then preprocessed and important features like Mel Frequency Cepstral Coefficients (MFCC) are extracted to capture the characteristics of speech. These features are used to train a machine learning model, such as a Multi-Layer Perceptron (MLP), which learns to classify emotions like happy, sad, angry, and neutral. After training, the model is integrated into a web-based application developed using Flask, with HTML and CSS used for the user interface. The system allows users to upload audio files or record their voice in real time, processes the input, and predicts the corresponding emotion, displaying the result along with emojis. This step-by-step methodology ensures an efficient and user-friendly system for real-time speech emotion recognition.

3.1 Proposed Workflow

The proposed workflow of the Speech Emotion Detector begins with the user providing input in the form of either an uploaded audio file or real-time voice recording through the web application. The input audio is then passed to the preprocessing stage, where the speech signal is cleaned, normalized, and prepared for analysis. After preprocessing, important features such as Mel Frequency Cepstral Coefficients (MFCC), pitch, and energy are extracted from the audio signal to represent the characteristics of the speech.These extracted features are then fed into a pre-trained machine learning model, such as a Multi-Layer Perceptron (MLP), which has been trained using a labeled emotional speech dataset. The model analyzes the input features and predicts the corresponding emotion category, such as happy, sad, angry, calm, or neutral. Finally, the predicted emotion is displayed on the web interface along with visual elements like emojis for better user understanding. This workflow ensures a smooth and efficient process from input acquisition to real-time emotion prediction, making the system practical and user-friendly.

3.2 Architectural Overview

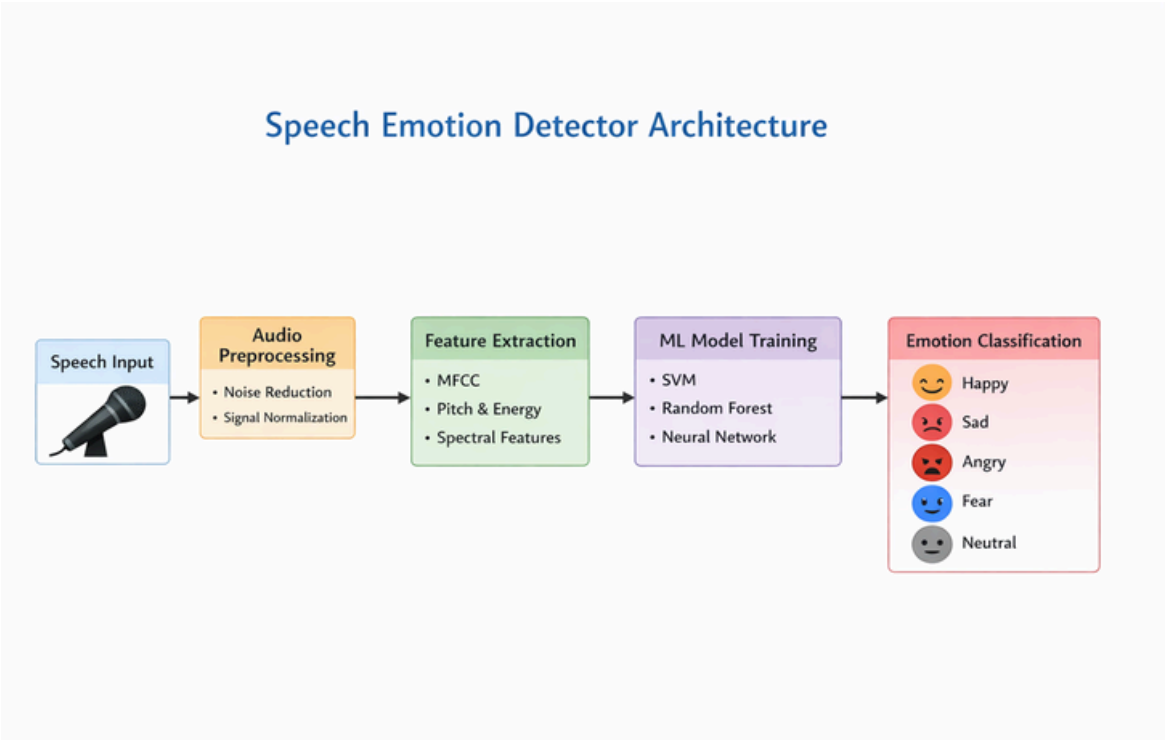

Fig 3.2.1 Architecture Diagram

The diagram illustrates the architecture of the Speech Emotion Detector showing the complete workflow from input to output. The process begins with speech input, where the user provides audio through a microphone or an audio file, which is then passed to the audio preprocessing stage for noise reduction and signal normalization to improve quality and consistency. Next, in the feature extraction stage, important characteristics such as MFCC, pitch, energy, and spectral features are extracted from the processed audio. These features are then used in the machine learning model training stage, where algorithms like SVM, Random Forest, and Neural Networks learn patterns associated with different emotions. Finally, in the emotion classification stage, the trained model predicts the emotion of the speech, such as happy, sad, angry, fear, or neutral, and displays the result, often with visual indicators like emojis, ensuring an efficient and accurate emotion detection process.

3.3 Hardware and Software Requirements

The Speech Emotion Detector requires both hardware and software components for its implementation. The hardware requirements include a computer or laptop with at least an Intel Core i3 processor, a minimum of 4 GB RAM (8 GB recommended for better performance), sufficient storage space to handle the dataset and project files, and a microphone for real-time voice recording. On the software side, the system is developed using Python as the programming language, with Flask as the web framework for building the application. Various libraries such as NumPy and Pandas are used for data handling, Librosa for audio processing and feature extraction, and Scikit-learn for training the machine learning model. Additionally, SoundDevice and SoundFile libraries are used for audio recording and processing. The user interface is designed using HTML and CSS, and the application can be run on operating systems like Windows, Linux, or macOS using development tools such as VS Code or PyCharm, and accessed through any modern web browser like Google Chrome.

3.4 System Integration and Working

The Speech Emotion Detector integrates machine learning, audio processing, and web technologies to provide an efficient and user-friendly platform for emotion recognition. The system is developed using Python for backend processing, where a trained machine learning model is used to predict emotions, and Flask is used to connect the backend with the frontend. The user interface is designed using HTML and CSS, allowing users to upload audio files or record their voice in real time. When the user provides input, the audio is sent to the server, where it undergoes preprocessing and feature extraction, such as MFCC. These features are then passed to the trained model, which analyzes the data and predicts the corresponding emotion. The result is sent back to the web interface and displayed along with emojis for better visualization. This integration ensures smooth communication between all components and enables fast and accurate emotion detection in real-time applications.

CHAPTER 4
RELEVANCE OF THE PROJECT

4.1 Why the Model Was Chosen

The Multi-Layer Perceptron (MLP) model was chosen for this project because it can effectively handle the complex and non-linear patterns present in speech data, which are difficult for simpler algorithms to capture. Since speech signals contain various variations in tone, pitch, and intensity, MLP, being a type of artificial neural network, is well-suited to learn these patterns from extracted features such as Mel Frequency Cepstral Coefficients (MFCC). It provides a good balance between accuracy and computational efficiency, making it suitable for real-time applications. Additionally, MLP is easy to implement using libraries like Scikit-learn and does not require very high computational resources compared to deep learning models. It performs well on medium-sized datasets like RAVDESS and can accurately classify multiple emotion categories. The model also generalizes well when properly trained, making it a practical and reliable choice for this project, especially for integration within a web-based system.

4.2 Comparison with Other Models

In speech emotion detector, various machine learning models can be used, each with its own advantages and limitations. Traditional models such as Support Vector Machines (SVM) provide good accuracy and are effective for high-dimensional data, but they require careful tuning and may not perform well with large datasets. k-Nearest Neighbors (KNN) is simple and easy to implement, but it becomes slow with large datasets and is sensitive to noise. Decision Trees and Random Forest models offer better interpretability and can handle complex data, but they may sometimes lead to overfitting if not properly managed. Deep learning models like Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN), including LSTM, can achieve higher accuracy by automatically learning features from raw audio data, but they require large datasets, high computational power, and longer training time.

Compared to these models, the Multi-Layer Perceptron (MLP) used in this project provides a balanced approach. It is capable of capturing non-linear patterns in speech data while being less computationally intensive than deep learning models. MLP is easier to implement, faster to train, and performs well on medium-sized datasets like RAVDESS. Although it may not reach the highest accuracy of advanced deep learning models, it offers a good trade-off between performance, simplicity, and efficiency, making it suitable for real-time web-based emotion detection applications.

Feature	**SVM (Support Vector Machine)**	**KNN(K-Nearest Neighbors)**	**MLP (Neural Network)**
Accuracy	High	Medium	High
Speed	Medium	Slow	Medium
Complexity	Medium	Low	High
Data Need	Medium	Low	Medium to High
Implementation	Moderate	Easy	Moderate

4.3 Advantages and Disadvantages

The Speech Emotion Detector offers several advantages, including the ability to automatically detect human emotions from speech, support for both real-time voice recording and audio file upload, and quick predictions using machine learning techniques. It enhances human-computer interaction by enabling systems to understand user emotions and has applications in areas such as mental health monitoring, customer service analysis, and virtual assistants. The web-based interface built using Flask, HTML, and CSS makes it user-friendly and easily accessible. However, the system also has some limitations, such as reduced accuracy in the presence of background noise or poor-quality audio, dependence on the quality and size of the training dataset, and difficulty in handling different accents and speaking styles.

Additionally, the use of standard datasets and feature extraction techniques helps maintain consistency and reliability in performance.

However, the system also has certain limitations. Its accuracy can be affected by background noise, low-quality recordings, and variations in speech such as different accents, tones, and speaking styles. The performance of the model largely depends on the quality and diversity of the training dataset, and limited data may reduce its ability to generalize in real-world scenarios. It may not fully capture complex or mixed human emotions, as emotions are often subtle and overlapping. Real-time processing might introduce slight delays depending on system performance and resource availability. Furthermore, achieving higher accuracy may require more advanced models and increased computational resources, which can add to the system complexity.

4.4 Summary of Relevance

The Speech Emotion Detector is highly relevant in today's technology-driven world, where there is an increasing need for intelligent systems that can understand human emotions. By combining machine learning and speech processing techniques, this project enables computers to interpret emotional cues from voice, making interactions more natural and effective. It has significant relevance in various real-world applications such as mental health monitoring, where emotional analysis can help in early detection of stress or depression, customer service systems for analyzing user satisfaction, and virtual assistants that can respond more intelligently based on user emotions. Additionally, the use of a web-based platform makes the system easily accessible and practical for everyday use. Overall, this project highlights the growing importance of emotion-aware systems and contributes to the development of more advanced, responsive, and human-centric technologies.

CHAPTER 5
MODULE DESCRIPTION

5.1 Data Collection Module

The Data Collection Module plays a fundamental role in the development of the Speech Emotion Detection System, as the overall performance of the model highly depends on the quality, quantity, and diversity of the dataset used. In this project, a standard emotional speech dataset such as RAVDESS is utilized, which contains professionally recorded audio samples representing various emotional states including happy, sad, angry, calm, fear, disgust, surprise, and neutral. These recordings are performed by multiple speakers with different tones, intensities, and speaking styles, which helps the model learn a wide range of emotional patterns. The dataset is well-structured and labeled, allowing for effective supervised learning. Each audio file follows a naming convention that encodes important information such as emotion type, intensity, and speaker identity, making it easier to organize and process the data efficiently. In addition to collecting the dataset, this module also involves organizing the audio files into directories and ensuring that only relevant .wav files are selected for processing.

5.2 Data Preprocessing Module

The Data Preprocessing Module is responsible for preparing the raw audio data for further analysis by improving its quality and consistency. Speech signals often contain noise, silence, and variations in recording conditions, which can negatively impact the performance of the machine learning model. This module performs several important operations such as loading audio files using libraries like Librosa, trimming unnecessary silence, reducing background noise, and normalizing the amplitude of the audio signals. These steps ensure that the input data is clean and standardized, which is essential for accurate feature extraction and model training.

Furthermore, the preprocessing module ensures that all audio samples have a uniform sampling rate and fixed duration. This is important because machine learning models require consistent input dimensions to function effectively.

In some cases, audio signals may be padded or truncated to maintain uniform length. By transforming raw and unstructured audio into a clean and consistent format, this module significantly enhances the efficiency of the system and improves the accuracy of emotion prediction. It acts as a crucial step that bridges the gap between raw data and meaningful analysis.

5.3 Feature Extraction Module

The Feature Extraction Module is one of the most critical components of the system, as it converts raw audio signals into numerical representations that can be understood by machine learning algorithms. In this project, Mel Frequency Cepstral Coefficients (MFCC) are used as the primary features because they effectively capture the frequency characteristics of human speech in a way that closely resembles human auditory perception. MFCC helps in identifying subtle variations in speech that are associated with different emotional states, making it highly suitable for emotion detection tasks.

In addition to MFCC, other features such as pitch, energy, spectral centroid, and zero-crossing rate may also be extracted to enhance the model's performance. These features provide additional information about the intensity, tone, and dynamics of the speech signal. The extracted features are then converted into feature vectors, which serve as input to the machine learning model. This module plays a vital role in bridging the gap between raw audio data and computational analysis, ensuring that the model receives meaningful and relevant information for accurate emotion classification.

5.4 Model Training Module

The Model Training Module is responsible for building an intelligent system capable of recognizing emotions from speech data. In this project, a Multi-Layer Perceptron (MLP) classifier is used due to its ability to handle complex and non-linear relationships in the data. The extracted feature vectors and corresponding emotion labels are divided into training and testing sets, allowing the model to learn from one portion of the data and validate its performance on another. During training, the model adjusts its internal parameters using optimization techniques to minimize prediction errors and improve accuracy.

The training process involves multiple iterations, during which the model learns to identify patterns associated with different emotions. Hyperparameters such as the number of hidden layers, neurons, and training iterations are carefully selected to achieve optimal performance. Once the model is trained and evaluated, it is saved as a file using the Pickle library, allowing it to be reused without retraining. This module is the core component of the system, as it enables the transformation of extracted features into meaningful predictions.

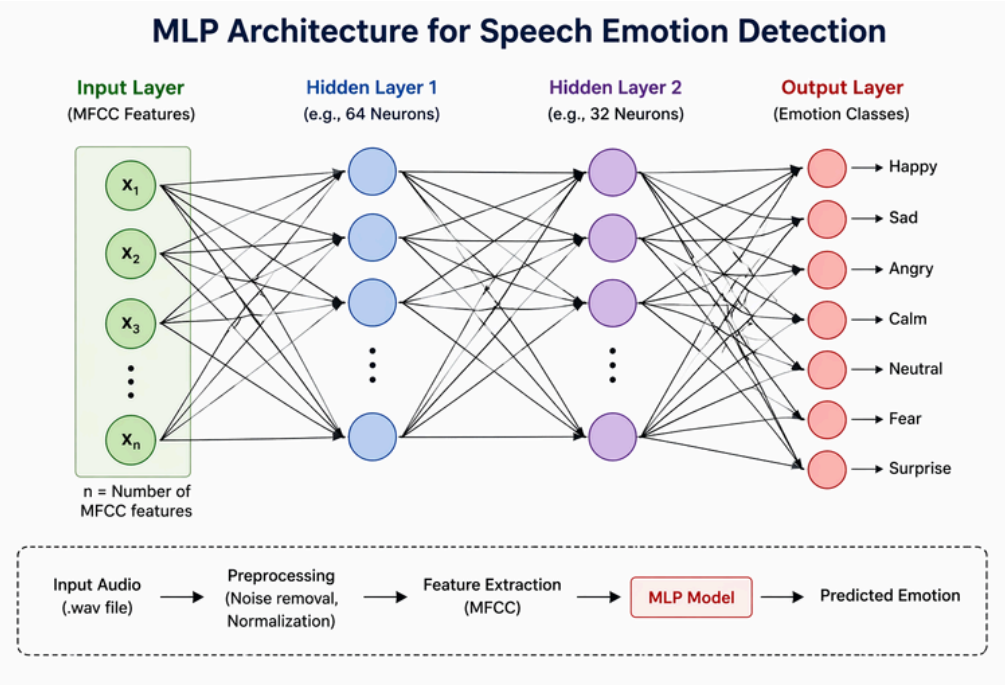

Fig 5.4.1 Multi-Layer Perceptron (MLP) Model

5.5 Prediction Module

The Prediction Module is responsible for identifying the emotional state of a speaker based on new audio input. When a user uploads an audio file or records their voice in real time, the system processes the input using the same preprocessing and feature extraction techniques applied during the training phase. This ensures consistency between training and testing conditions, which is essential for accurate predictions. The processed feature vector is then passed to the trained MLP model, which analyzes the data and predicts the corresponding emotion.

The prediction process is designed to be fast and efficient, enabling real-time or near real-time emotion detection. The system ensures low latency so that users receive immediate feedback. This module plays a crucial role in delivering the final output of the system and is responsible for converting learned patterns into actionable insights. It ensures that the system functions effectively in practical scenarios, providing accurate emotion predictions for both uploaded and live audio inputs.

5.6 Web Interface Module

The Web Interface Module provides a platform through which users can interact with the system. It is developed using Flask for backend processing and HTML and CSS for designing the frontend interface. The interface includes features such as audio file upload, real-time voice recording, and displaying prediction results. JavaScript may also be used to enable recording functionalities such as start and stop recording. The goal of this module is to create a simple, intuitive, and user-friendly interface that enhances accessibility. The results of the emotion prediction are displayed clearly on the web page using text and emojis, making it easy for users to understand the output. The interface is designed to be responsive and compatible with different devices and browsers. This module ensures smooth communication between the user and the backend system, allowing users to interact with the application without needing technical knowledge. It plays a key role in improving user experience and making the system practical for real-world use.

5.7 Evaluation Module

The Evaluation Module is used to assess the performance and effectiveness of the trained machine learning model. It involves calculating various metrics such as accuracy, precision, recall, and confusion matrix to evaluate how well the model predicts different emotions. This module helps in identifying strengths and weaknesses in the model and provides insights for improvement.

By analyzing evaluation results, developers can fine-tune the model, adjust parameters, and improve overall performance. This module ensures that the system is reliable and capable of handling real-world data.

CHAPTER 6
RESULT AND DISCUSSION

6.1 Performance Analysis

The performance of the Speech Emotion Detector is evaluated based on how accurately and efficiently the model predicts emotions from speech data. The trained Multi-Layer Perceptron (MLP) model is tested using a portion of the dataset that was not used during training, ensuring unbiased evaluation. Key performance metrics such as accuracy, precision, recall, and F1-score are used to measure the effectiveness of the model. Accuracy indicates the overall correctness of predictions, while precision and recall provide insights into how well the model identifies specific emotion classes. The confusion matrix is also used to visualize the comparison between actual and predicted emotions, helping to identify misclassification patterns.

The system achieves satisfactory performance when trained on a well-balanced dataset like RAVDESS, with accuracy typically ranging between moderate to high depending on feature quality and model tuning. The use of MFCC features significantly improves classification performance by capturing important speech characteristics. However, the system's performance may vary when tested on real-world audio due to background noise, variations in speech, and differences in recording conditions. Despite these challenges, the model provides reliable results with relatively low computational cost, making it suitable for real-time applications. Further improvements in performance can be achieved by increasing dataset size, using advanced feature extraction techniques, or implementing deep learning models.

6.2 User Feedback

The user feedback for the Speech Emotion Detectior has been generally positive, highlighting its effectiveness, ease of use, and practical functionality. Users find the web-based interface simple and intuitive, allowing them to easily upload audio files or record their voice in real time without requiring technical knowledge. The system's ability to quickly process input and display predicted emotions along with emojis makes it engaging and user-friendly.

Many users appreciate the real-time response and the clarity of the results, which helps them understand emotional states more effectively. The integration of machine learning with a web interface is also seen as a strong aspect, making the system accessible and convenient for various applications such as basic emotional analysis, communication tools, and experimental learning.

However, users have also provided constructive feedback regarding certain limitations of the system. Some have observed that prediction accuracy can vary depending on audio quality, background noise, and differences in speech patterns such as accents, tone, and speaking speed. In real-time recording, occasional delays or inconsistencies may occur based on system performance. Additionally, users have suggested improving the system by including more emotional categories, enhancing model accuracy, and adding features such as confidence scores or graphical outputs. Overall, the feedback indicates that while the system performs well and meets its core objectives, there is scope for further enhancement to make it more robust, accurate, and adaptable to real-world conditions.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Summary of Outcomes

The Speech Emotion Detector successfully demonstrates the ability to analyze and classify human emotions from speech using machine learning techniques. The project achieves its primary objective by developing a functional web-based application that allows users to upload audio files or record their voice in real time and receive emotion predictions. By utilizing feature extraction techniques such as MFCC and implementing a Multi-Layer Perceptron (MLP) model, the system is able to identify emotions like happy, sad, angry, calm, and neutral with satisfactory accuracy. The integration of backend processing using Python and Flask with a simple and interactive frontend ensures smooth user interaction and quick response time.

The outcomes of the project highlight the effectiveness of combining audio signal processing with machine learning for emotion recognition tasks. The system performs well under controlled conditions and provides reliable predictions, demonstrating its practical applicability in areas such as human-computer interaction, mental health analysis, and customer feedback systems. Additionally, the modular design of the system makes it easy to maintain and extend for future improvements. Although some limitations exist, such as sensitivity to noise and variation in speech patterns, the project successfully establishes a strong foundation for further development and enhancement of emotion-aware intelligent systems.

7.2 Future Scope and Enhancements

The Speech Emotion Detector offers a wide range of future scope and enhancement opportunities that can significantly improve its performance, usability, and real-world applicability. One of the key areas for improvement is the adoption of advanced deep learning models such as Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and Long Short-Term Memory (LSTM) networks, which are capable of capturing complex temporal and spatial patterns in speech data more effectively than traditional machine learning models. Additionally, increasing the size and diversity of the training dataset by including multiple languages, accents, age groups, and real-world noisy environments can greatly enhance the model’s ability to generalize and provide more accurate predictions.

Incorporating advanced feature extraction techniques and combining multiple features can further improve the system's performance.

Another important area of enhancement is improving the robustness and efficiency of the system in real-time scenarios. Implementing advanced noise reduction and speech enhancement techniques can help the system perform better in environments with background noise. The prediction speed can also be optimized to ensure faster response times, making the system more suitable for real-time applications. Furthermore, the system can be extended to support continuous emotion tracking in live conversations instead of analyzing only short audio clips. This would enable more dynamic and context-aware emotion detection, which is highly useful in applications like virtual assistants and customer interaction systems.

From a user experience perspective, the web application can be enhanced by adding more interactive and informative features. For example, displaying confidence scores, graphical representations such as charts, and historical analysis of emotions can provide deeper insights to users. The interface can be further improved with better design, animations, and responsiveness to make it more engaging. Integration with mobile applications or deployment on cloud platforms can increase accessibility and scalability, allowing users to access the system from anywhere. Security features such as user authentication and data privacy measures can also be added to make the system more reliable and suitable for real-world deployment.

In terms of practical applications, the system can be integrated into various domains such as mental health monitoring, where it can help in early detection of stress, anxiety, or depression by analyzing speech patterns over time. It can also be used in customer service systems to analyze customer emotions and improve service quality, or in educational platforms to understand student engagement. Additionally, the system can be combined with other modalities such as facial expression recognition and text sentiment analysis to create a multimodal emotion detection system, which would provide more accurate and comprehensive results. Overall, these future enhancements can transform the current system into a more powerful, intelligent, and widely applicable emotion recognition solution.

APPENDICES

APPENDIX A

SOURCE CODE

```
from flask import Flask, render_template, request
import librosa
import numpy as np
import pickle
import sounddevice as sd
import soundfile as sf

app = Flask(__name__)

model = pickle.load(open("emotion_model.pkl","rb"))
emotion_emoji = {
"happy":"    ",
"sad":"    ",
"angry":"    ",
"calm":"    ",
"neutral":"    ",
"fear":"    ",
"disgust":"    ",
"surprise":"    "
}
def extract_feature(file):
    audio, sr = librosa.load(file, duration=3)
    mfcc = np.mean(librosa.feature.mfcc(y=audio, sr=sr, n_mfcc=40).T, axis=0)
    return mfcc.reshape(1,-1)
@app.route("/")
def index():
    return render_template("index.html")
@app.route("/predict", methods=["POST"])
def predict():
    file = request.files["file"]
```

```
path = "uploads/" + file.filename

    file.save(path)
    feature = extract_feature(path)
    prediction = model.predict(feature)[0]
    emoji = emotion_emoji.get(prediction,"")
    result = f"Predicted Emotion : {prediction} {emoji}"
    return render_template("index.html", prediction=result)

@app.route("/record")
def record():
    duration = 4
    fs = 44100
    print("Recording...")
    audio = sd.rec(int(duration*fs), samplerate=fs, channels=1)
    sd.wait()

    file = "recorded.wav"
    sf.write(file, audio, fs)
    feature = extract_feature(file)
    prediction = model.predict(feature)[0]
    emoji = emotion_emoji.get(prediction,"")
    result = f"Predicted Emotion : {prediction} {emoji}"
    return render_template("index.html", prediction=result)

if __name__ == "__main__":
    app.run(debug=True)
```

```
//train model
import librosa
import numpy as np
import os
import pickle
from sklearn.model_selection import train_test_split
from sklearn.neural_network import MLPClassifier
dataset_path = r"C:\Users\ELCOT\Downloads\emo\dataset"
emotions = {
    "01":"neutral",
    "02":"calm",
    "03":"happy",
    "04":"sad",
    "05":"angry",
    "06":"fear",
    "07":"disgust",
    "08":"surprise"
}
X=[]
y=[]

def extract_feature(file):
    audio, sr = librosa.load(file, duration=3)
        mfcc = np.mean(librosa.feature.mfcc(y=audio, sr=sr, n_mfcc=40).T,
axis=0)
    return mfcc
for root,dirs,files in os.walk(dataset_path):
    for file in files:
        if file.endswith(".wav"):
```

```
 emotion = file.split("-")[2]

            if emotion in emotions:
                path = os.path.join(root,file)
                feature = extract_feature(path)
                X.append(feature)
                y.append(emotions[emotion])

X=np.array(X)

X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2)

model=MLPClassifier(hidden_layer_sizes=(300,),max_iter=500)

model.fit(X_train,y_train)

pickle.dump(model,open("emotion_model.pkl","wb"))

print("Model trained and saved")
```

APPENDIX B – Screenshots

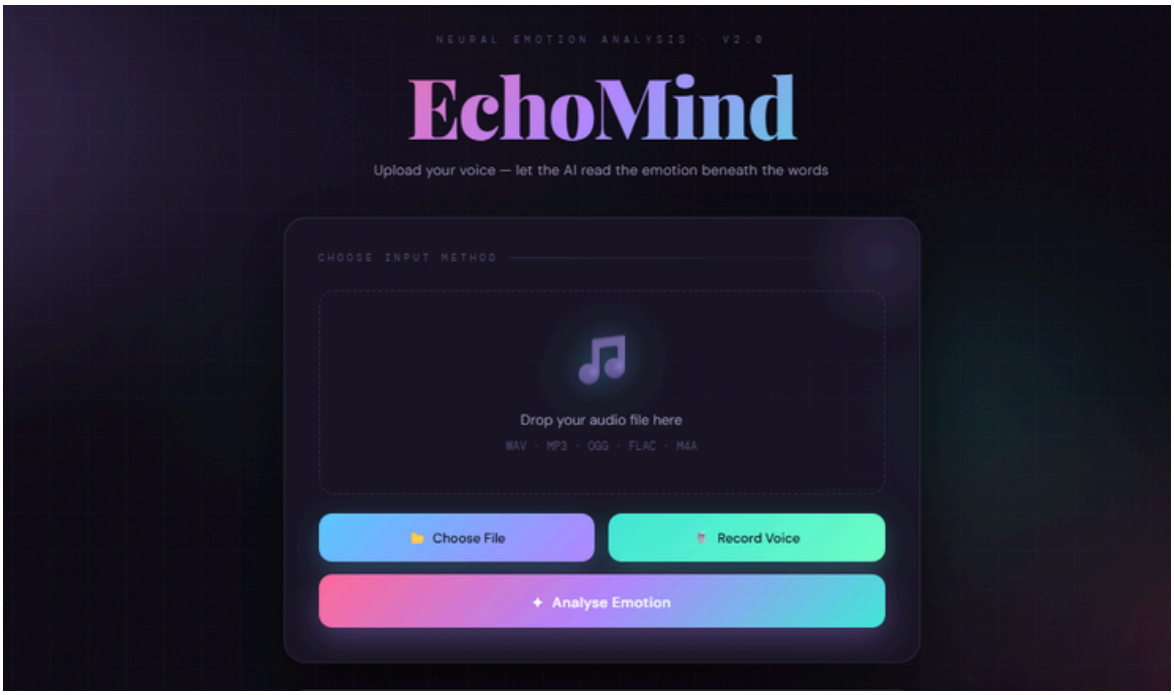

Figure B.1 Home Page

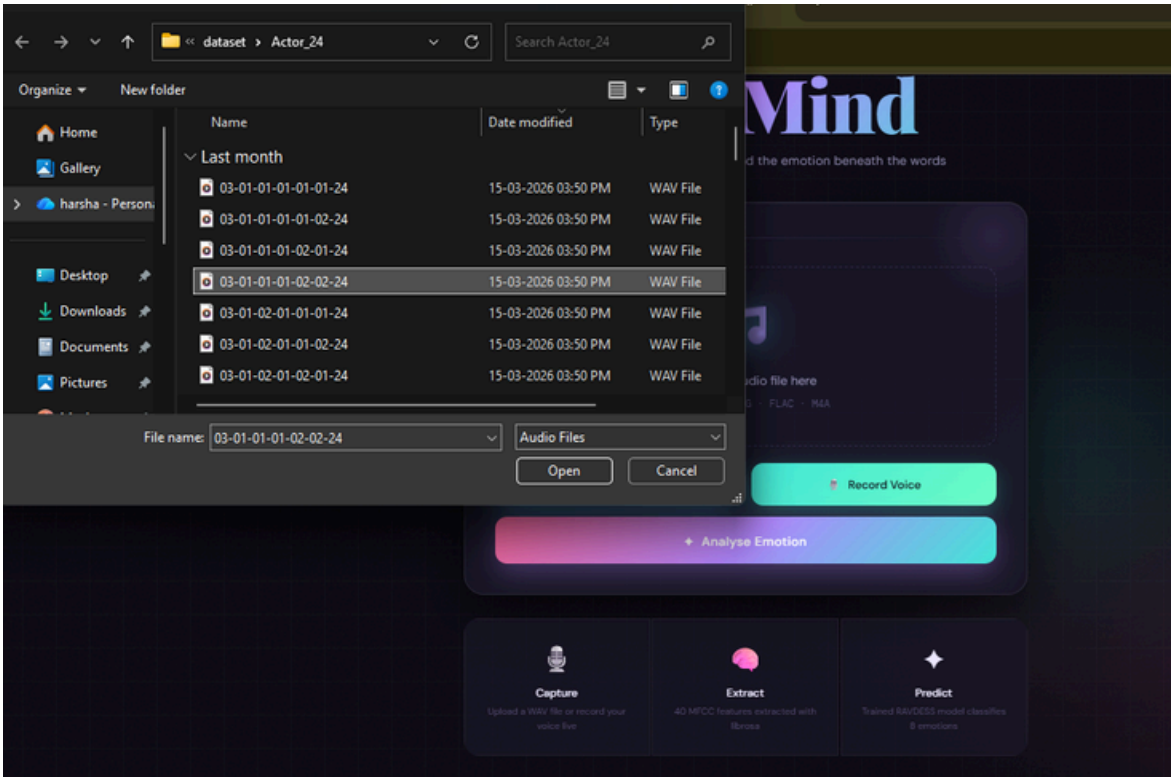

Fig B.2: File Uploading Screen

Figure B.3 Prediction Results Output

REFERENCES

[1] Livingstone, S. R., & Russo, F. A. (2018). The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS). PLoS ONE.

[2] Librosa Documentation. (n.d.). Retrieved from https://librosa.org

[3] Scikit-learn Documentation. (n.d.). Retrieved from https://scikit-learn.org

[4] Flask Documentation. (n.d.). Retrieved from https://flask.palletsprojects.com

[5] Speech and Language Processing, Jurafsky, D., & Martin, J. H. (2009). Pearson Education.

[6] El Ayadi, M., Kamel, M. S., & Karray, F. (2011). Survey on speech emotion recognition: Features, classification schemes, and databases. Pattern Recognition Journal.

[7] IEEE Xplore Digital Library – Research papers on speech emotion recognition.

[8] Kaggle – Speech Emotion datasets and implementations.